

BE AHEAD IN 5G

With The Right T&M System

5G networks deliver faster and more reliable communications. These open the doors to exciting new connections to the Internet of Things (IoT) networks, autonomous driving, broadband fixed wireless and faster video viewing. However, delivering 5G involves end-to-end test, precision measurement and deep network visibility solutions to ensure smooth operation across the ecosystem



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5G, the next generation of mobile networks, is going to change the way we communicate. It introduces a paradigm shift towards a user- and application-centric technology framework, with the goal of flexibly supporting three important use case families. Enhanced mobile broadband (eMBB) focuses on supporting the ever-increasing end user data rate and system capacity, while massive machine-type communications (mMTC) targets the cost-efficient

and robust connection of billions of devices without overloading the network. With ultra-reliable low latency communications (URLLC), 5G opens up an entirely new use case family by supporting new requirements from vertical industries such as autonomous driving for the automotive industry, remote surgery for e-health and cloud robotics for Industry 4.0. Finally, increased capacity will significantly reduce or eliminate congestion issues and facilitate machine-to-

machine (M2M) communications at a scale that will take system interconnectivity to a new level.

All these above-mentioned use case families require new technology components—from components and chipsets to assembled user devices and base stations—that challenge product development. Moreover, these also come with their own technology challenges including the spectrum shift to higher frequencies and the need for advanced antenna arrays to enable